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A Calculus of Fidelity-Graded Translations — Supplementary Appendix

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This supplementary appendix accompanies the paper and ships with its anonymized artifact (this document lives at `paper/appendix/`). Numbered references are to the paper’s July 2026 snapshot: Definition 3.5 (Faithfulness), Definition 3.6 (Support), Theorem 3.7 (Pasting), Corollary 3.8 (Localization), Proposition 3.9 (Determinism and caching), Proposition 4.2 (Weakest link), Definition 4.6 (Coverage), and Proposition 4.7 (Ratchet).

A Proofs

A.1 Pasting (Theorem 3.7)

We spell out the chain, including the domain side conditions elided in the paper’s proof sketch. Let $p \in \text{dom}(P_2 \circ P_1)$, i.e. $p \in \text{dom}(I_A) \cap \text{dom}(T_1)$, $q := T_1(p) \in \text{dom}(T_2)$, $r := T_2(q)$, $I_C(r)$ defined, $\Lambda_2(I_C(r)) \in \text{dom}(\Lambda_1)$, and (from $p \in \text{dom}(P_1)$, $q \in \text{dom}(P_2)$) $I_B(q)$ defined with $I_B(q) \in \text{dom}(\Lambda_1)$ and $I_C(r) \in \text{dom}(\Lambda_2)$.

1. By faithfulness of P_1 at p and $\pi \subseteq \pi_1$, $\pi(I_A(p)) = \pi(\Lambda_1(I_B(q)))$.
2. By faithfulness of P_2 at q : $\pi_2(I_B(q)) = \pi_2(\Lambda_2(I_C(r)))$.
3. Both $I_B(q)$ and $\Lambda_2(I_C(r))$ lie in $\text{dom}(\Lambda_1)$ (the first by $p \in \text{dom}(P_1)$, the second by $p \in \text{dom}(P_2 \circ P_1)$). By $(\pi_2 \Rightarrow \pi)$ -support applied to step (2): $\pi(\Lambda_1(I_B(q))) = \pi(\Lambda_1(\Lambda_2(I_C(r))))$.
4. Chaining (1) and (3): $\pi(I_A(p)) = \pi(\Lambda_1(\Lambda_2(I_C(r))))$, which is faithfulness of $P_2 \circ P_1$ at p w.r.t. π . $\square$

Necessity of support. Without the support condition (Definition 3.6) the theorem is false: let $F_B = \{g, h\}$, $\pi_2 = \{g\}$, and let Λ_1 copy field h into a π_1 -kept source field. Take P_2 faithful (it preserves g) but with $\Lambda_2(I_C(r))$ differing from $I_B(q)$ on h — permitted, since $h \notin \pi_2$. The outer rectangle then disagrees on the copied field although both inner squares commute.

A.2 Composed keep-set, and step granularity

For fieldwise, step-aligned Λ_1 (each kept source field f computed per step as $f = \phi_f(\{b.g \mid g \in \text{deps}(f)\})$), $(\pi_2 \Rightarrow \pi)$ -support holds iff $\text{deps}(f) \subseteq \pi_2$ for every $f \in \pi$; hence the largest admissible π is $\text{keep} = \{f \in \pi_1 \mid \text{deps}(f) \subseteq \pi_2\}$ and support for it is checkable syntactically from the dependency sets.

When Λ_1 retimes — one source step corresponds to a window of target steps, as in C statements over ISA instructions — model Λ_1 as a monotone window map w from source step indices to intervals of target step indices plus a fieldwise read within each window. Support then requires $\text{deps}(f) \subseteq \pi_2$ for reads *anywhere in the window*, and additionally that the window map itself be determined by π_2 -observables (e.g. by

a kept program counter), since window boundaries influence the projected output. Both conditions are again syntactic given w and the dependency sets.

A.3 Localization (Corollary 3.8)

Contrapositive of the pasting theorem (Theorem 3.7) for two hops. For n hops, induct on the route: if the outer rectangle over hops $1..n$ fails at p but hop 1’s square passes at p , then by the two-hop case applied to (hop 1, rectangle over hops $2..n$) the rectangle over $2..n$ fails at $T_1(p)$; recurse. Termination yields a least failing hop i ; within it, the square oracle’s comparison yields the least failing step and a witnessing field. $\square$

A.4 Weakest link (Proposition 4.2)

Soundness: if every hop is universal on its fragment, iterate the pasting theorem (Theorem 3.7) over all p in the composite fragment. If some hop is perrun, the same iteration is available exactly at programs whose per-hop images passed the oracle — the per-run set of the route. If some hop is only replay, purity composes (Proposition 3.9) but no faithfulness statement is available for the route beyond it. Optimality: a route with a perrun hop entails no universal statement (take a defect outside the checked set); a route with a replay hop entails no faithfulness at all (take a pure but wrong translator). $\square$

A.5 Ratchet (Proposition 4.7)

Let $I' \sqsupseteq I$ and let $p \in \text{dom}(I)$. Every quantity in the faithfulness square (Definition 3.5) evaluated at p reads only I -values on $\text{dom}(I)$ -points, which I' preserves; hence the verdict at p is unchanged. Coverage monotonicity: cov’s per-construct conjunction can only gain (new constructs enter dom) and never lose (old verdicts preserved). Composition of extensions is an extension. For the failure half: a non-extension update changes some $I(p)$, which can flip the verdict at p for any pair whose square consumed it — justifying the versioned-event rule. $\square$

B Construct inventories and probe corpora

Per-language construct inventories (the yardsticks of the paper’s Definition 4.6) and the probe corpora behind every coverage figure in the paper’s §6 are enumerated in the artifact — the per-language inventory modules (`gurdy/languages/*/inventory.py`), the per-pair inventories (`gurdy/pairs/*/inventory.py`), and the measured coverage data (`paper/results/data/`);

they are spec-derived: the RISC-V RV64IMC opcode inventory, the full 144-opcode EVM inventory, the 256-opcode eBPF inventory slice, the Wasm numeric/parametric/control

inventory, the OpenSMILES construct list, the Python-subset AST allow-list, and the two hubs' operator inventories.